

Section 1: Multiple Choice Questions:

1/ C

2/ ~~D~~ D

3/ ~~D~~ D

4/ C

5/ A

6/ C

7/ B

8/ C

9/ A

10/ B

11/ B

12/ C

13/ B

14/ C

15/ D

16/ C

17/ A

18/ C

19/ B

20/ C

Section 2:

- 21). Supervised Learning is a machine learning model that ~~that~~ train computers ~~with~~ by feeding labeled data for it to do learning.
- 25) List out any two major applications of data science:
- + Real Estate price prediction
 - + Medical sector: Heart disease detection/analysis.

Section 2 (continue)

- 31) K in K means algorithm represent number of clusters.
- 34) + ~~Picture 1~~: classifier 1: the model classifies well by splitting the two class with enough length between each class
- + Classifier 2: the ~~model~~ SVM model, classifies the data poorly where it has several datapoints in between the parallel lines (classifier lines)
- + Classifier 3: the SVM model classifies the data into two classes but the ~~length~~ distance between the parallel lines are still small.
- 28). Data pre processing is the process to prepare data before it gets into the analysis or modelling phase which could mean the data is biased and leads to rubbish model. Data pre processing includes: Data profiling, cleansing, transformation, reduction, extraction and normalization.

- 22). Model M1 has 99% accuracy for training dataset and 30% accuracy for testing dataset.
+ Comment: Model M1 still needs to do more optimization by changing some dependant ~~parameter~~ ~~and~~.
- 23). Regression analysis is a technique to identify the the relationship between features and to output the continuous value prediction.
- 26).
+ Structure data: are data with proper that comes with proper rows & column (table format),
+ ~~semi-structured~~ Ex: database data; excel format; csv.
+ semi structured data = ~~is~~ is datatype that structure in various format such as XML; JSON; HTML...
- 27). We can find out an outlier in the Data by:
+ visualization
+ calculating distances ~~from the~~ between data points
+ sorting technique
- 29). Data normalization is required in K-NN because K-NN algorithm is a classification algorithm that need to group/categorised data when data is large it leads to K-NN to provide inaccurate class and it consume more time to compute the distances.

24). Elucidate any two challenge of Machine Learning:

- + Data pre processing: data must be clean and properly segregate the content. If the data model is fed with rubbish data, it will also ~~pre~~output rubbish.

- + Features selection: identifying the relationships between independent and dependent variables that would optimizly affect the performance of the model.

30). The continuous attributes are discretized by using the regression methods for instance: Linear Regression,

33). Gradient descent is ~~an algorithm to~~ machine learning ^{Unsupervised} ~~classification~~ technique which is to find relationship between ~~parent-child data~~ predicted data.

32). Hyperparameters of machine learning algorithm:

- + Accuracy
- + Confusion matrix
- + Evaluation
- + Time complexity